Amara Nwosu

Earth Observation Data Scientist · Nairobi, Kenya
amara@example.com · Sample resume: the person, companies and projects are fictional.

I turn satellite imagery into crop, flood and land cover maps that county officers, insurers and farmers' groups can act on, from segmentation models to the pipelines that run them every pass.

EXPERIENCE

Earth Observation Data Scientist, Mbuni Earth Analytics — Nairobi · Mar 2023 to Present

- Lead modelling for Rainline, which maps flood extent and crop condition for 14 county governments and two crop insurers.
- Built the radar flood segmentation model that cut pass-to-map time from two days to a median of 3.5 hours.
- Run a field validation program with 30 enumerators, turning their GPS checks into a held-out test set that every release must beat.
- Mentor three junior analysts on model evaluation, label quality and writing maps up for non-specialists.

Data Scientist, Remote Sensing, Greenrow Agritech — Nairobi · Jun 2020 to Feb 2023

- Built crop type and field boundary models for smallholder plots under one hectare across western Kenya.
- Designed the drought stress index behind a farm advisory SMS service used by about 90,000 farmers.

GIS and Remote Sensing Analyst, Lakeshore Water Trust — Kisumu · Jan 2018 to May 2020

- Mapped 24 years of shoreline and papyrus wetland change around the lake for a conservation program.
- Trained county staff to run the water mapping workflow themselves in open source GIS tools.

FLAGSHIP: RAINLINE

Every satellite pass becomes a flood and crop map a county officer can use the same day. **Ingest**: Pulls each new radar and optical scene, masks cloud and shadow, and tiles it onto one 10 m grid. **Segment**: A U-Net with a temporal encoder labels water, cropland, built-up and bare ground for every pixel. **Change**: Compares the new mask with a seasonal baseline to find fresh flooding and crop loss, not just wet fields. **Alert**: Sends county officers a map and a short written summary, ranked by people and hectares affected.

Covers 14 counties, about 210,000 km² · Median 3.5 hours from satellite pass to alert · Flood extent F1 of 0.91 against field surveys

SELECTED PROJECTS

- **Flood Extent from Radar** (2024). Segments open and vegetated flood water from radar scenes, through cloud, during the long rains. 0.91 flood F1 vs field surveys
- **Smallholder Crop Type Mapping** (2023). Classifies maize, beans, sorghum, tea, sugarcane and grassland on plots smaller than one hectare. 84% accuracy over 6 classes
- **Field Boundary Delineation** (2022). Draws the edges of small farm fields from 3 m imagery so crop maps can be read per field. 0.81 boundary F1
- **Drought Stress Early Warning** (2022). Forecasts vegetation condition six weeks ahead from greenness, rainfall and soil moisture time series. 6 weeks forecast lead time
- **Settlement Growth Tracker** (2024). Finds new roofs and roads between yearly 50 cm image mosaics for county planning teams. 92% precision on new structures
- **Rift Basin Land Cover Map** (2023). An annual 10-class land cover map at 10 m for a river basin authority. 88% overall accuracy
- **Wetland Change Atlas** (2020). Twenty-four years of shoreline and papyrus wetland change around the lake, mapped year by year. 24 yrs of yearly maps

PUBLICATIONS AND TALKS

- Same-Day Flood Mapping for County Response with Radar Segmentation. ERSC-2025, Equatorial Remote Sensing Conference, Cape Town. ERSC Proceedings, 2025. **Best Paper Award.**
- Temporal Attention for Crop Type Mapping on Smallholder Plots. JAEI, Journal article. Journal of Applied Earth Imaging, 2024.
- Spending a Label Budget Wisely: Land Cover in Data-Poor Regions. Open Map Week 2024, Invited talk, Accra. Open Map Week, 2024.
- Twenty-Four Years of Wetland Loss on a Great Lake Shore. SIWO-2021, Symposium on Inland Water Observation, Lisbon. SIWO Proceedings, 2021.

EDUCATION

MSc, Geospatial Data Science, Baobab Coast University — Mombasa · Sep 2016 to Dec 2017

BSc, Geomatic Engineering, Baobab Coast University — Mombasa · Sep 2011 to Jul 2015

SKILLS

Remote sensing: Radar (SAR) processing, Optical time series, Cloud masking, Change detection, Accuracy assessment, Field validation
Machine learning: PyTorch, U-Net and segmentation, Temporal attention, Siamese networks, scikit-learn, LightGBM, Active learning
Geospatial data: GDAL, rasterio, xarray, Dask, PostGIS, QGIS, Cloud optimised rasters, STAC catalogues
Engineering: Python, SQL, FastAPI, Airflow, Docker, TypeScript, Git

CERTIFICATIONS

Advanced Radar Image Processing, Geodata Academy (Mar 2024) · Cloud Native Geospatial Pipelines, Open Raster Institute (Aug 2023) · Deep Learning for Imagery, Lumen ML School (Nov 2021) · Registered Geomatic Engineer, Board of Survey Professionals (Jun 2017)